

CLASS TOPIC

Foundations, data and descriptive statistics

What this class fixes

- Know the central question of the topic.
- Use the correct notation and units.
- Avoid mechanical calculation without interpretation.

W01**Statistics**

Clean formulas · aligned layout ·
corrected terminology



CLASS TOPIC

Foundations, data and descriptive statistics

Economic question first

- Start from a real decision: compare, describe, forecast or test.
- Translate the question into variables before using formulas.
- Define the population, sample and measurement unit.

FORMULA

$$\bar{x} = (1/n) \sum x_i$$

median = middle ordered value

$$\text{range} = \max(x) - \min(x)$$



CLASS TOPIC

Foundations, data and descriptive statistics

Key definitions

- Population: full group of interest.
- Sample: observed part of the population.
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CLASS TOPIC

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CLASS TOPIC

Foundations, data and descriptive statistics

Mini example

- Write the data clearly.
- Substitute values step by step.
- Round only at the final reporting stage.

01 Data: 4, 7, 8, 9, 12

02 Step 1: compute the needed statistic.

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FORMULA

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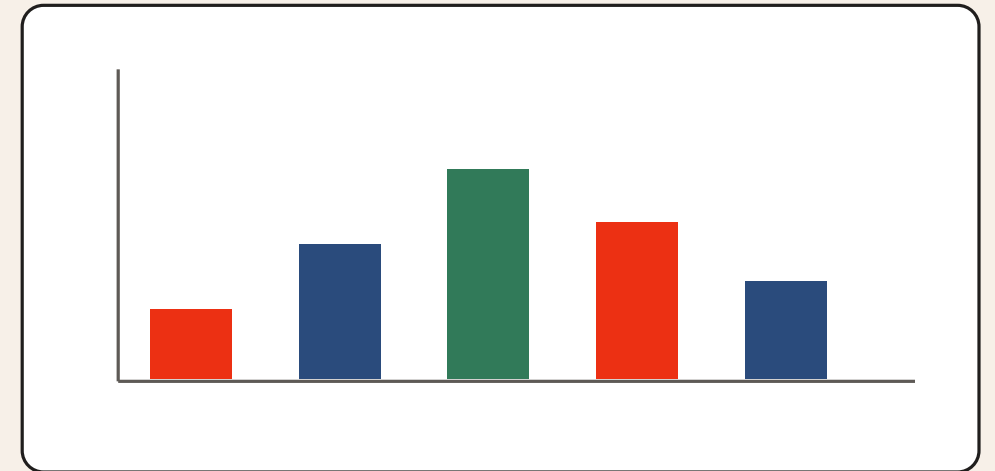
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CLASS TOPIC

Foundations, data and descriptive statistics

Visual intuition

- A graph should answer a question, not decorate the slide.
- Always label axes and units.
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**FORMULA**

$$\bar{x} = (1/n) \sum x_i$$

CLASS TOPIC

Foundations, data and descriptive statistics

Second formula box

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FORMULA 2

$$s^2 = \sum (x_i - \bar{x})^2 / (n - 1)$$

$$s = \sqrt{s^2}$$

$$CV = s / \bar{x}$$



CLASS TOPIC

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Common mistakes

- Mixing sample and population symbols.
- Using a formula without checking conditions.
- Reporting a number without units or interpretation.

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CLASS TOPIC

Foundations, data and descriptive statistics

R / spreadsheet check

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R / EXCEL

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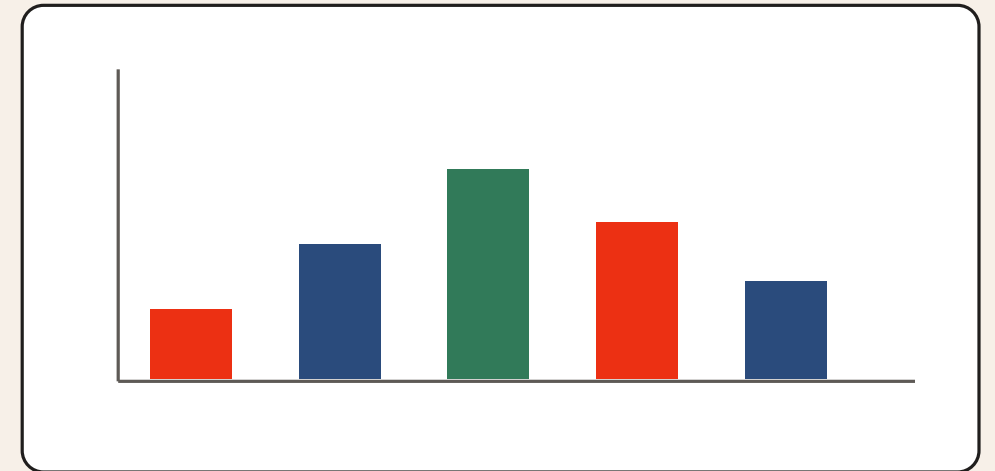
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CLASS TOPIC

Foundations, data and descriptive statistics

Interpretation sentence

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CLASS TOPIC

Foundations, data and descriptive statistics

Conditions and assumptions

- Check whether the method fits the data type.
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CLASS TOPIC

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Class exercise

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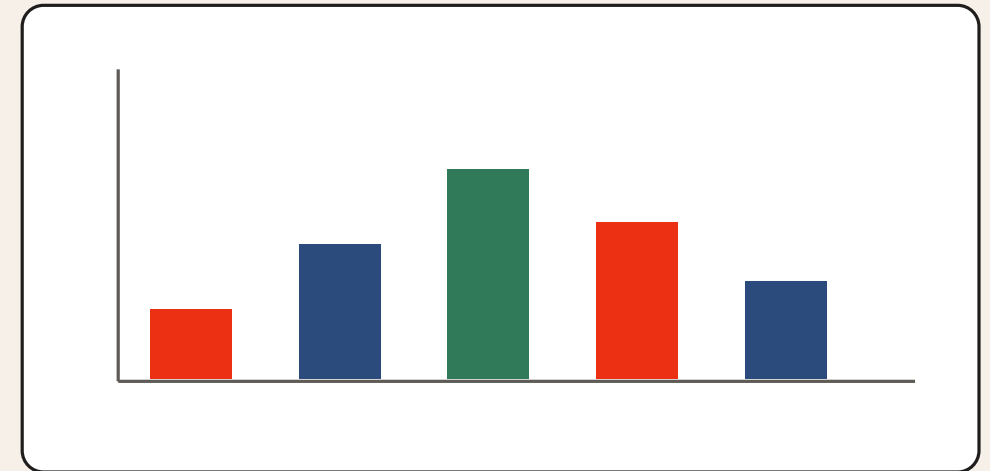


CLASS TOPIC

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How to report

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relative frequency = frequency / n

cumulative frequency = running total



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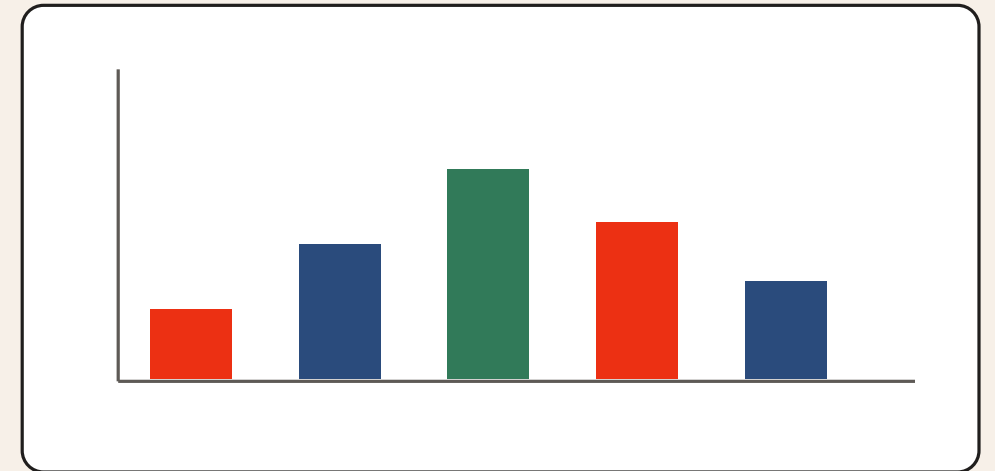
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FORMULA 2

$$Z_i = (x_i - \bar{x}) / s$$

$$\text{IQR} = Q_3 - Q_1$$

$$\text{outlier if } x < Q_1 - 1.5\text{IQR} \text{ or } x > Q_3 + 1.5\text{IQR}$$



CLASS TOPIC

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Common mistakes

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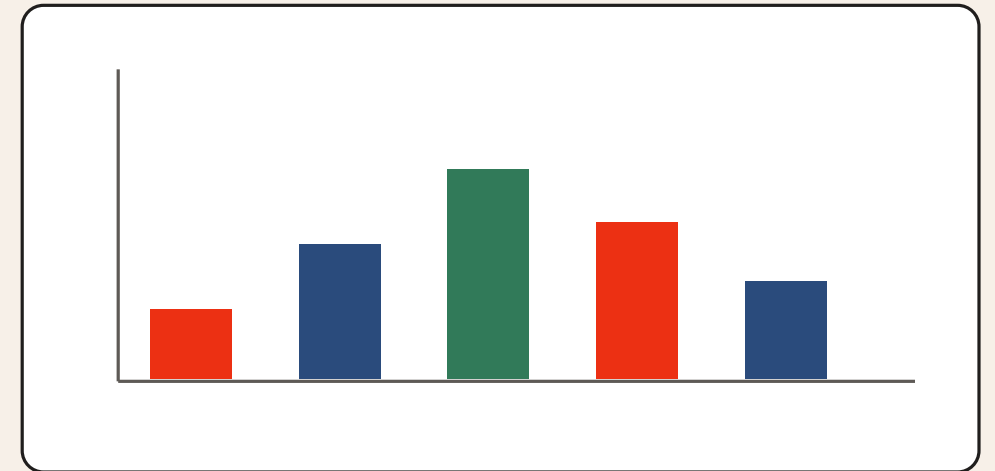
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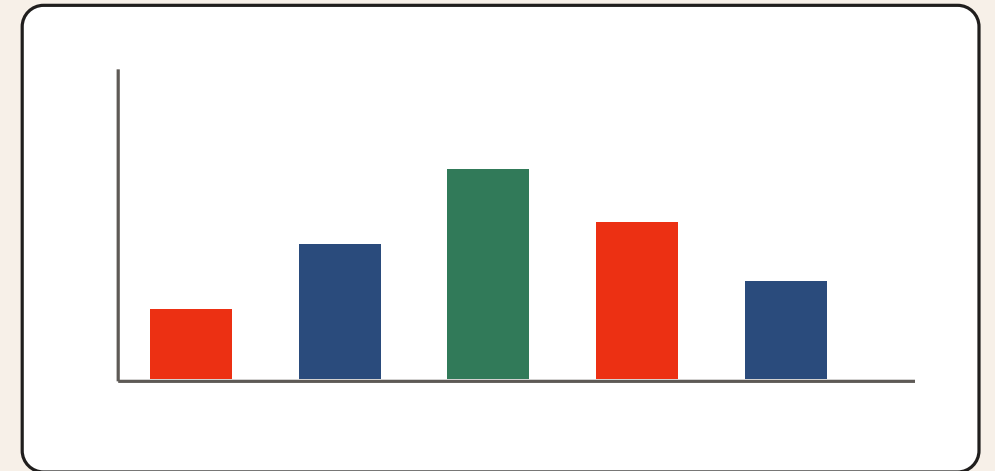


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CLASS TOPIC

Probability, conditional probability and Bayes

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W03**Statistics**

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FORMULA

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A^c) = 1 - P(A)$$



CLASS TOPIC

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CLASS TOPIC

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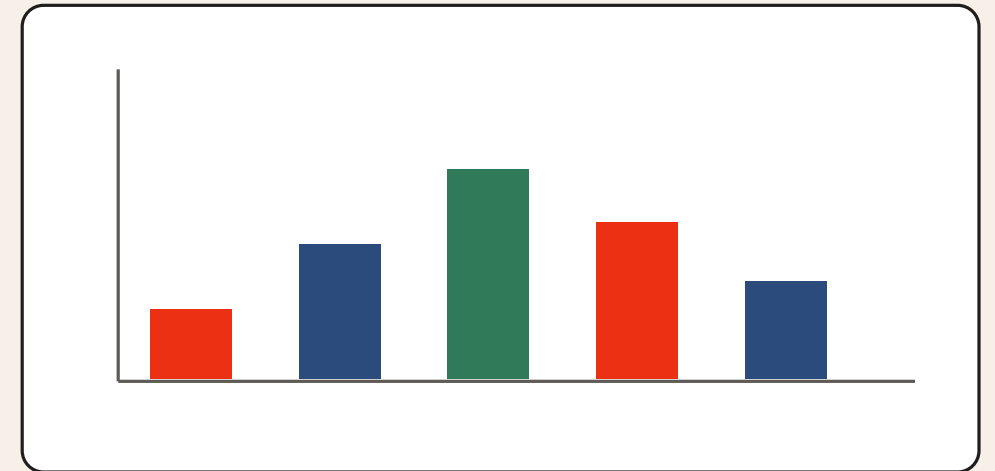
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CLASS TOPIC

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Visual intuition

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FORMULA 2

$$P(A \mid B) = P(A \cap B) / P(B)$$

$$P(A \mid B) = P(B \mid A)P(A) / P(B)$$



CLASS TOPIC

Probability, conditional probability and Bayes

Common mistakes

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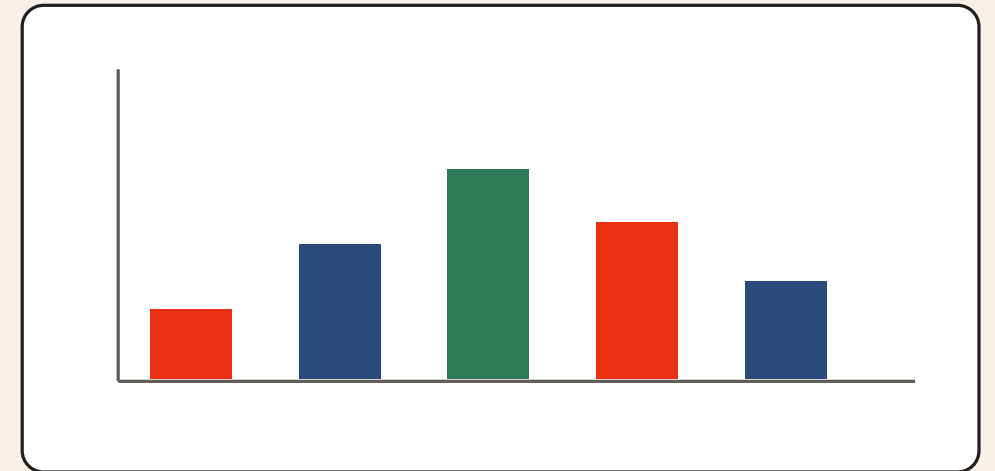
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CLASS TOPIC

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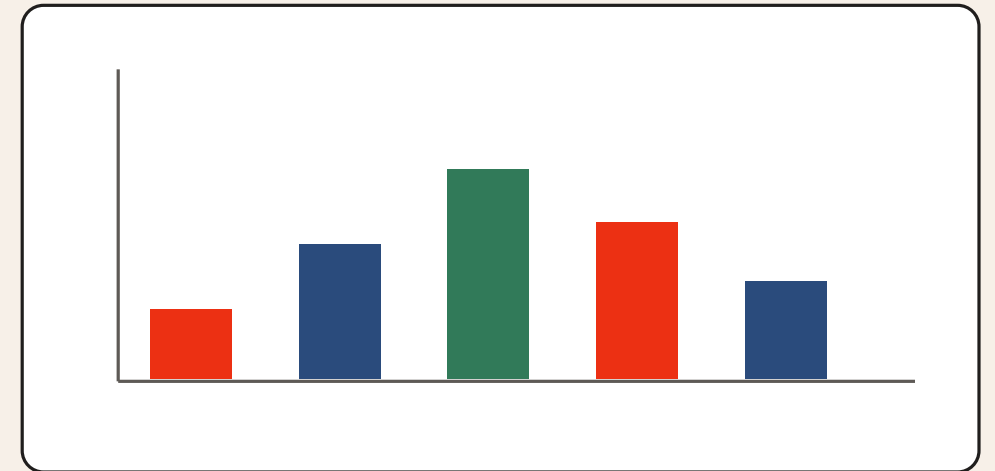


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CLASS TOPIC

Random variables and discrete distributions

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W04**Statistics**

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CLASS TOPIC

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FORMULA

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$$\text{Var}(X) = E[(X - \mu)^2] = \sum (x - \mu)^2 p(x)$$



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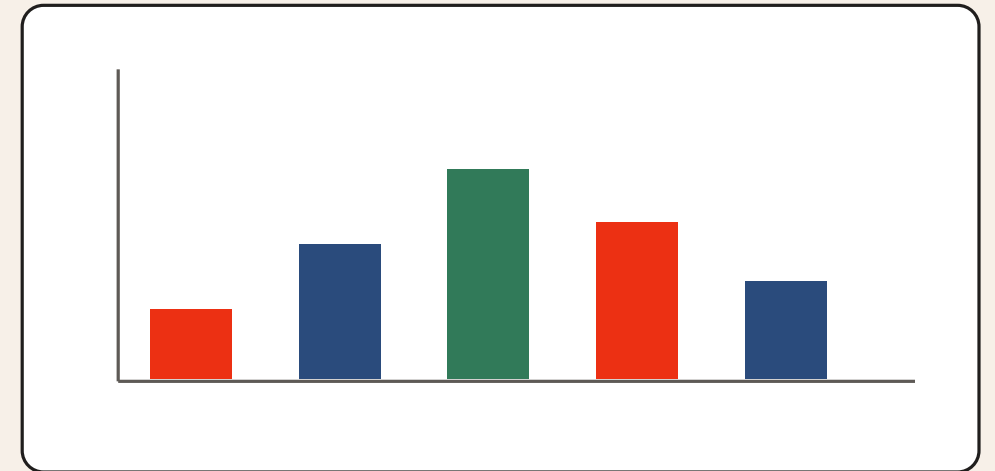
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FORMULA 2

Binomial: $P(X = k) = C(n, k)p^k(1 - p)^{n-k}$
 $E(X) = np$, $Var(X) = np(1 - p)$



CLASS TOPIC

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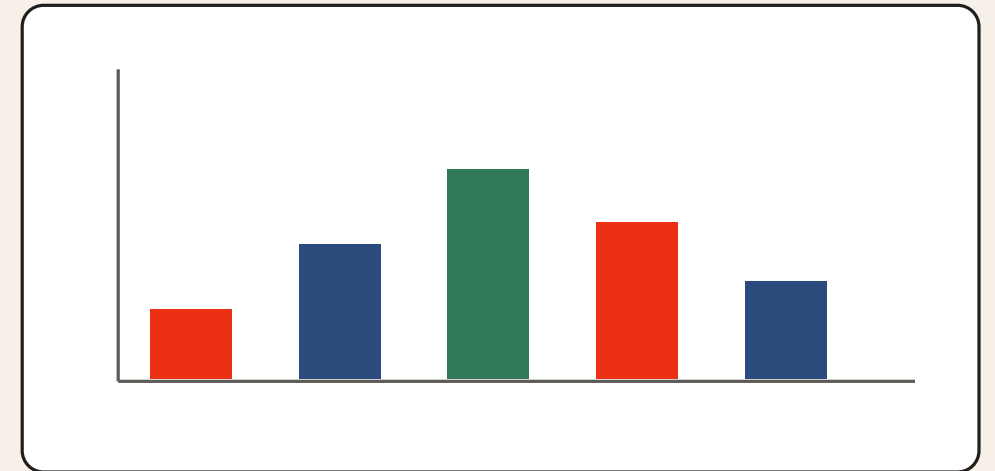
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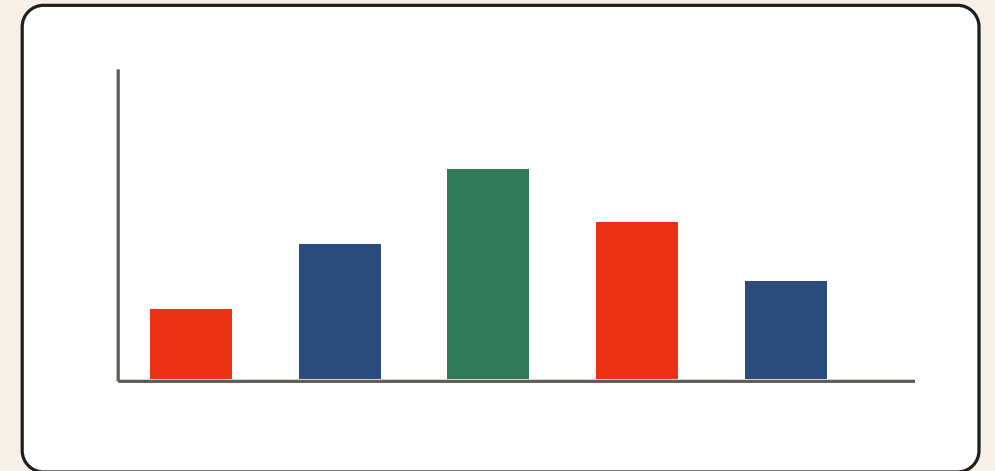


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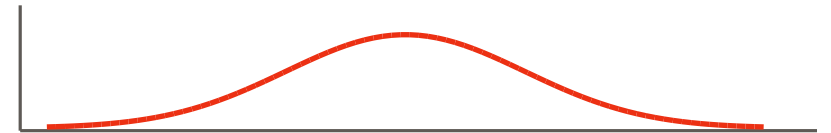
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FORMULA

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Total area under $f(x)$ equals 1



CLASS TOPIC

Continuous and normal distributions

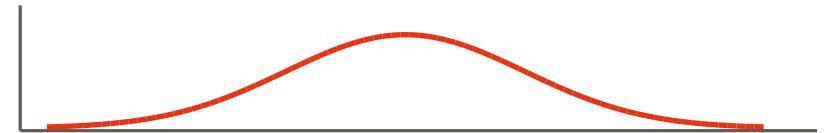
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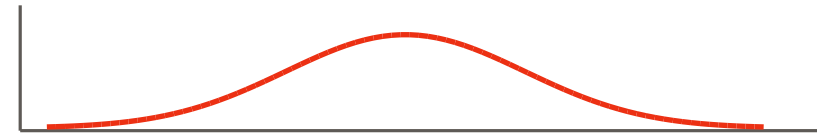
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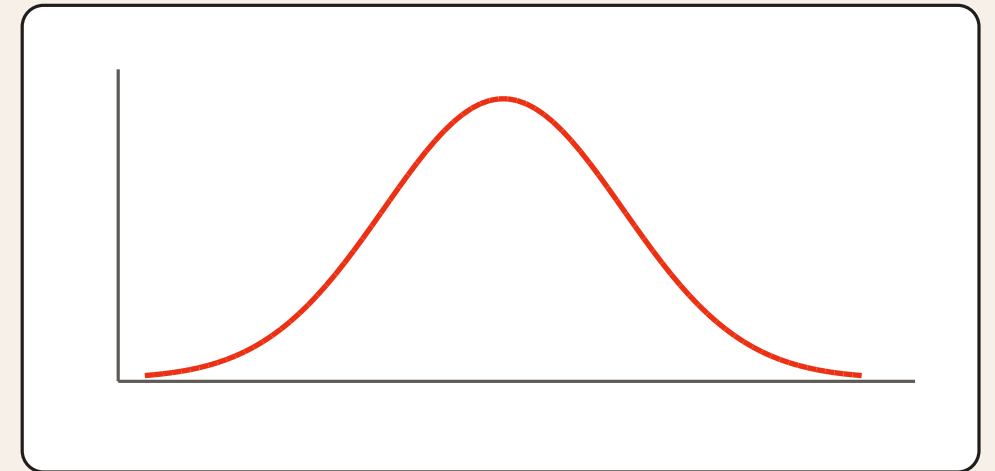
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CLASS TOPIC

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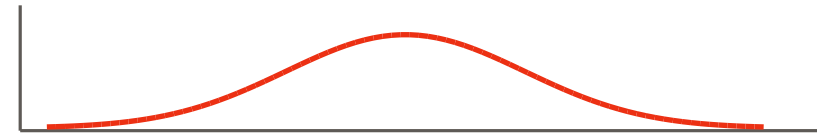
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FORMULA 2

$$Z = (X - \mu) / \sigma$$

$$X \sim N(\mu, \sigma^2)$$

$$P(X \leq x) = P(Z \leq (x - \mu) / \sigma)$$



CLASS TOPIC

Continuous and normal distributions

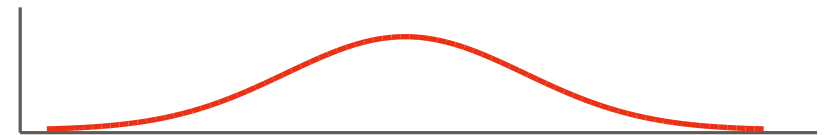
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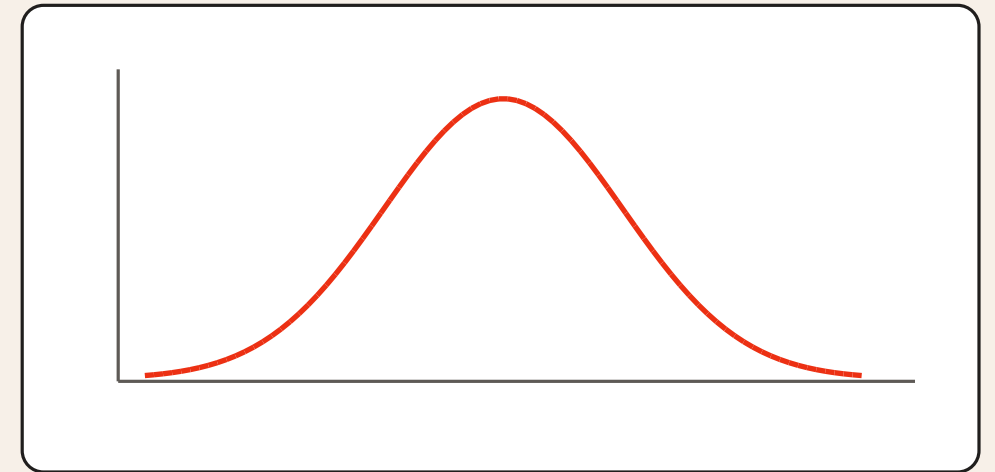
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CLASS TOPIC

Continuous and normal distributions

Interpretation sentence

- Do not stop at the number.
- Explain what the result means in the original context.
- Use cautious language when evidence is limited.

**FORMULA**

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

CLASS TOPIC

Continuous and normal distributions

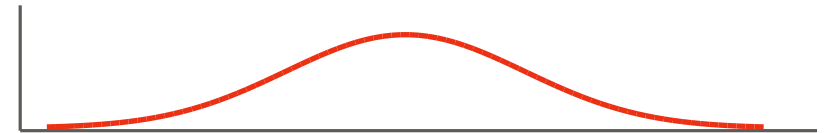
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CLASS TOPIC

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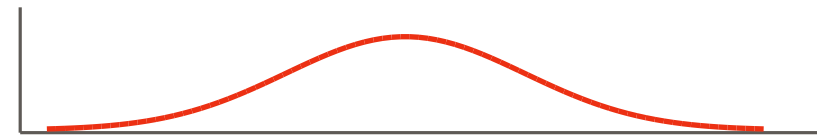
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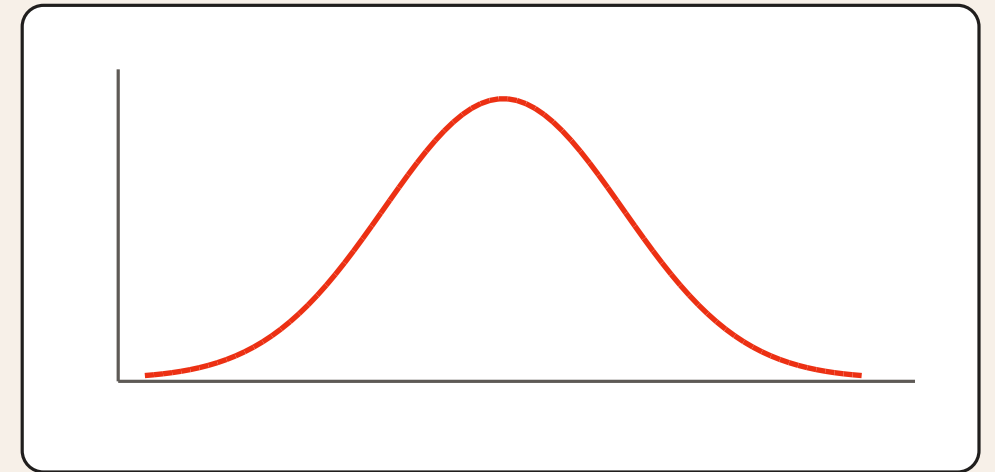


CLASS TOPIC

Continuous and normal distributions

How to report

- Report method, estimate, uncertainty and interpretation.
- Use tables only when they make comparison easier.
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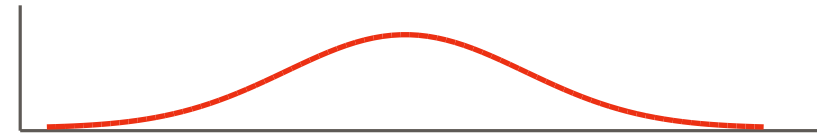
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FORMULA

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CLASS TOPIC

Joint distributions, covariance and correlation

What this class fixes

- Know the central question of the topic.
- Use the correct notation and units.
- Avoid mechanical calculation without interpretation.

W06**Statistics**

Clean formulas · aligned layout ·
corrected terminology



CLASS TOPIC

Joint distributions, covariance and correlation

Economic question first

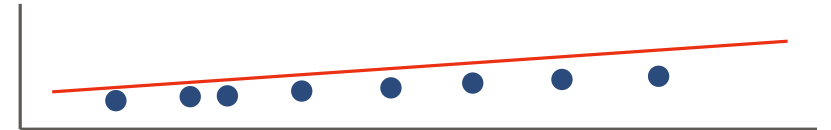
- Start from a real decision: compare, describe, forecast or test.
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FORMULA

$$\text{Cov}(X,Y) = E[(X-\mu_x)(Y-\mu_y)]$$

sample cov =

$$\Sigma(x_i - \bar{x})(y_i - \bar{y}) / (n-1)$$



CLASS TOPIC

Joint distributions, covariance and correlation

Key definitions

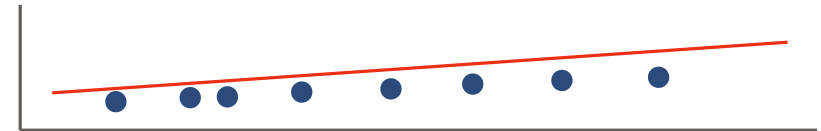
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FORMULA

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CLASS TOPIC

Joint distributions, covariance and correlation

Main formula box

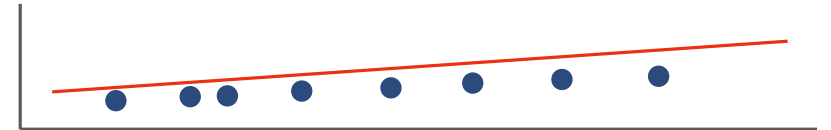
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CLASS TOPIC

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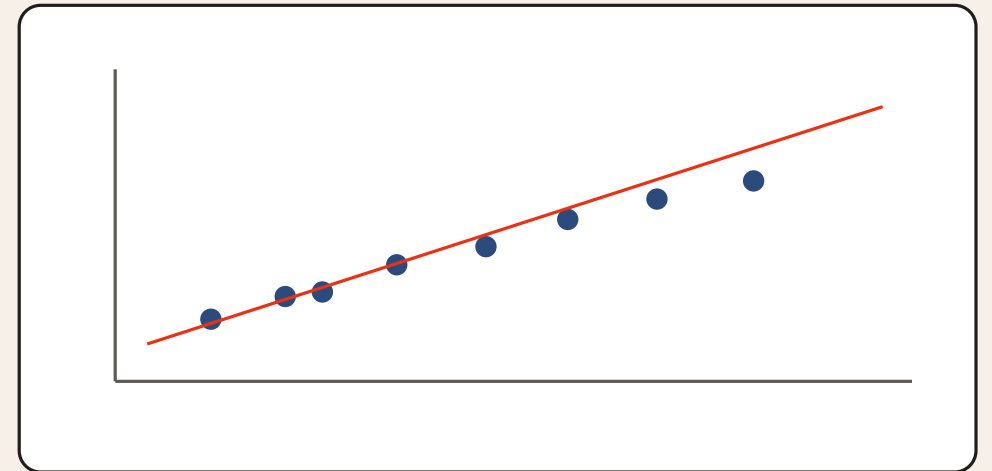
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CLASS TOPIC

Joint distributions, covariance and correlation

Visual intuition

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FORMULA

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CLASS TOPIC

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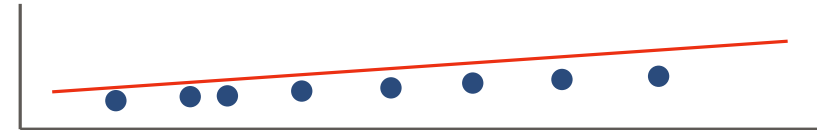
Second formula box

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FORMULA 2

$$\rho = \text{Cov}(X, Y) / (\sigma_X \sigma_Y)$$

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{[\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2]}}$$



CLASS TOPIC

Joint distributions, covariance and correlation

Common mistakes

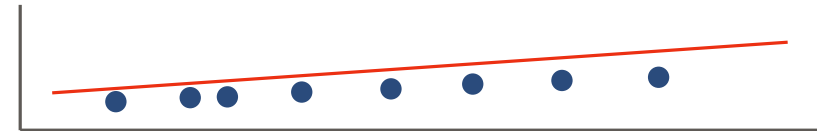
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CLASS TOPIC

Joint distributions, covariance and correlation

R / spreadsheet check

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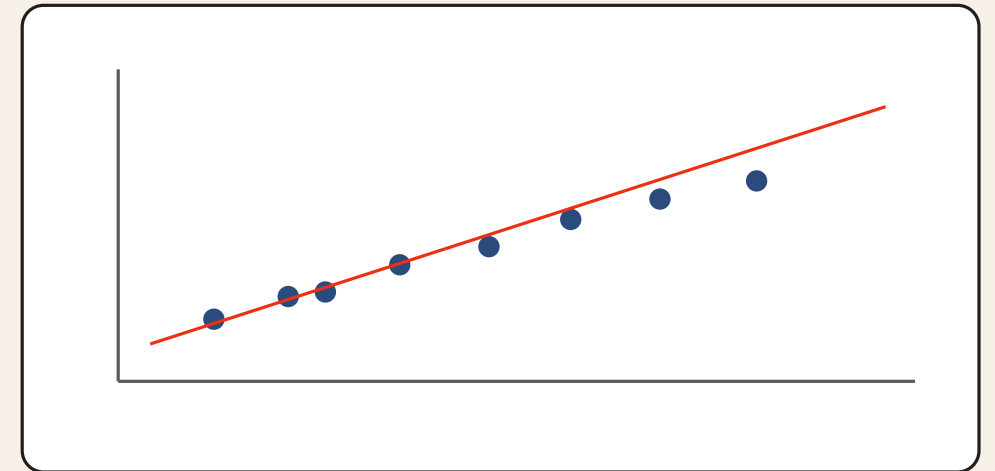
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CLASS TOPIC

Joint distributions, covariance and correlation

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CLASS TOPIC

Joint distributions, covariance and correlation

Conditions and assumptions

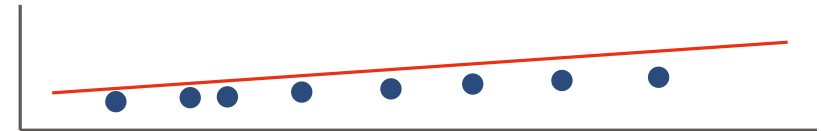
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CLASS TOPIC

Joint distributions, covariance and correlation

Class exercise

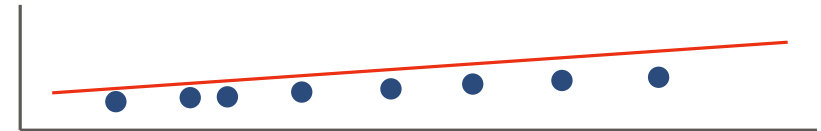
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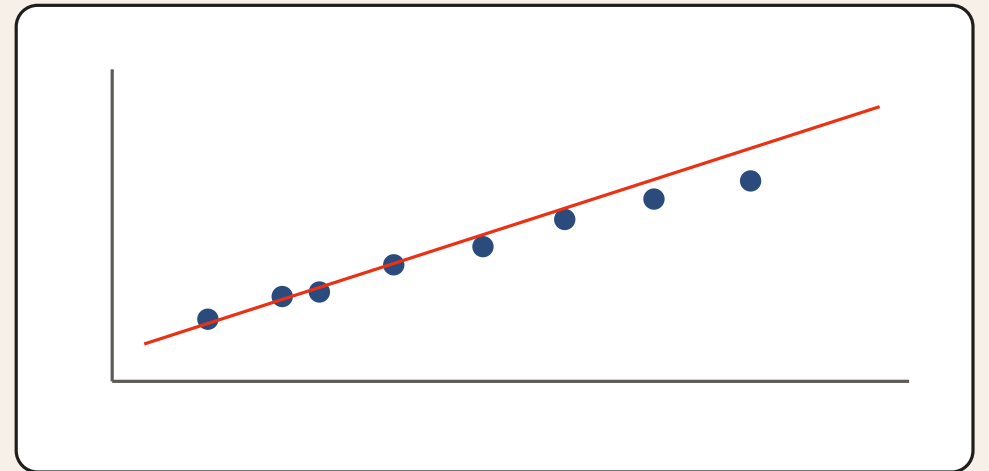


CLASS TOPIC

Joint distributions, covariance and correlation

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CLASS TOPIC

Joint distributions, covariance and correlation

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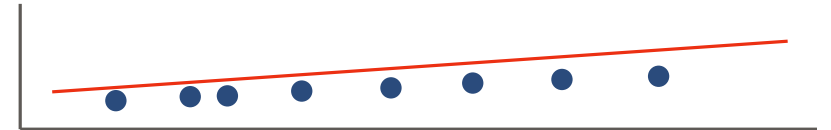
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CLASS TOPIC

Sampling distributions and the central limit theorem

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W07**Statistics**

Clean formulas · aligned layout ·
corrected terminology



CLASS TOPIC

Sampling distributions and the central limit theorem

Economic question first

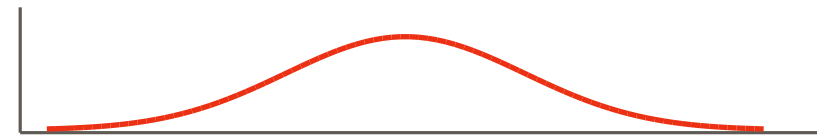
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FORMULA

$$E(\bar{X}) = \mu$$

$$SE(\bar{X}) = \sigma/\sqrt{n}$$

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CLASS TOPIC

Sampling distributions and the central limit theorem

Key definitions

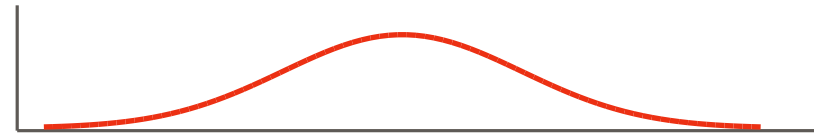
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CLASS TOPIC

Sampling distributions and the central limit theorem

Main formula box

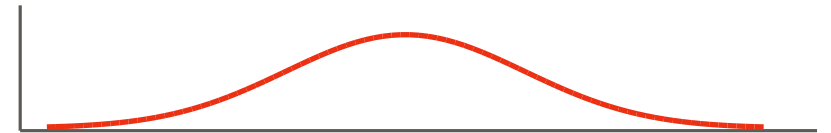
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CLASS TOPIC

Sampling distributions and the central limit theorem

Mini example

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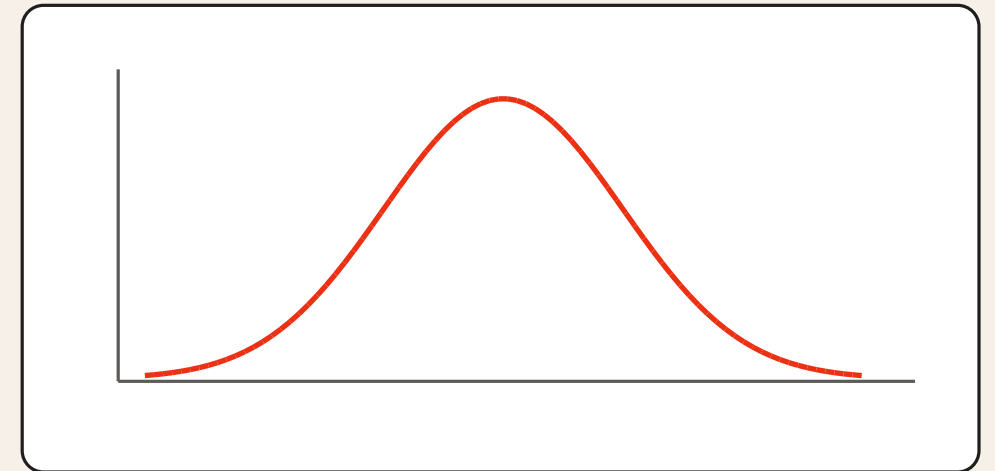
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CLASS TOPIC

Sampling distributions and the central limit theorem

Visual intuition

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CLASS TOPIC

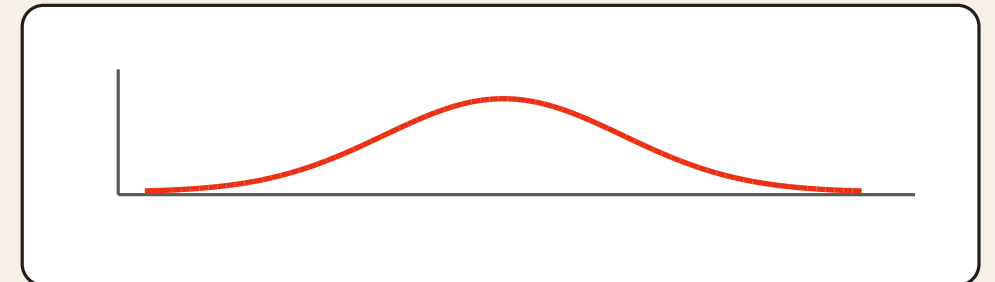
Sampling distributions and the central limit theorem

Second formula box

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FORMULA 2

$$\text{CLT: } \bar{X} \approx N(\mu, \sigma^2/n) \text{ for large } n$$
$$z = (\bar{x} - \mu)/(\sigma/\sqrt{n})$$



CLASS TOPIC

Sampling distributions and the central limit theorem

Common mistakes

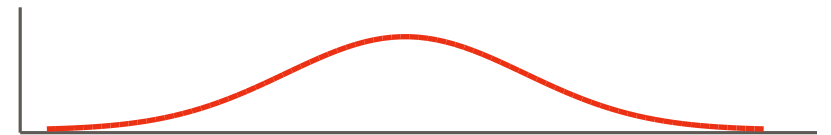
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CLASS TOPIC

Sampling distributions and the central limit theorem

R / spreadsheet check

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R / EXCEL

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FORMULA

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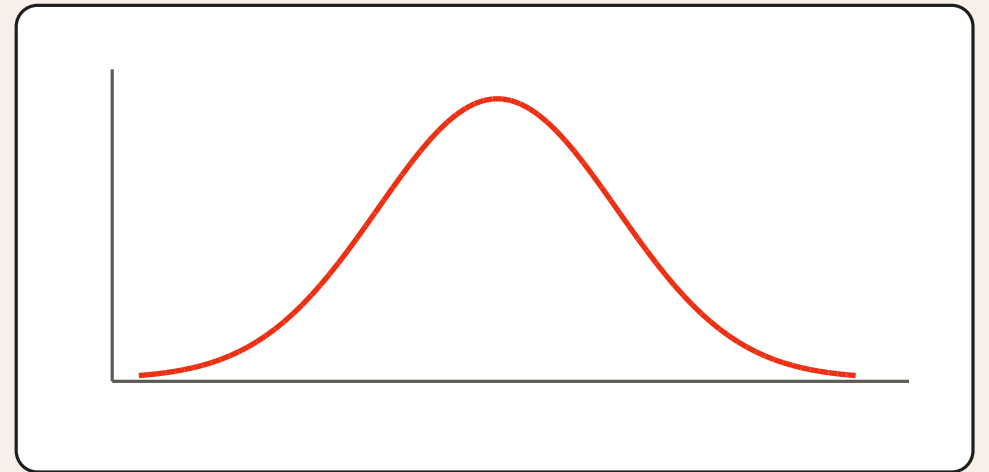
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CLASS TOPIC

Sampling distributions and the central limit theorem

Interpretation sentence

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CLASS TOPIC

Sampling distributions and the central limit theorem

Conditions and assumptions

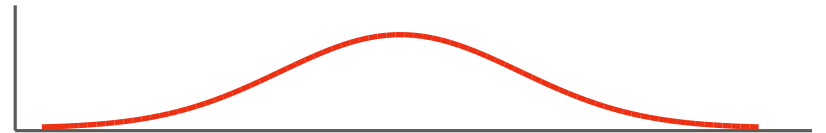
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CLASS TOPIC

Sampling distributions and the central limit theorem

Class exercise

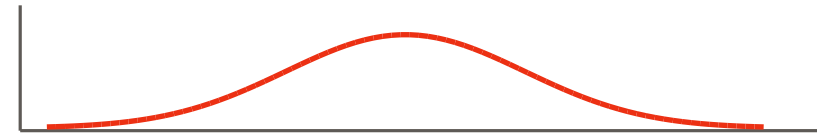
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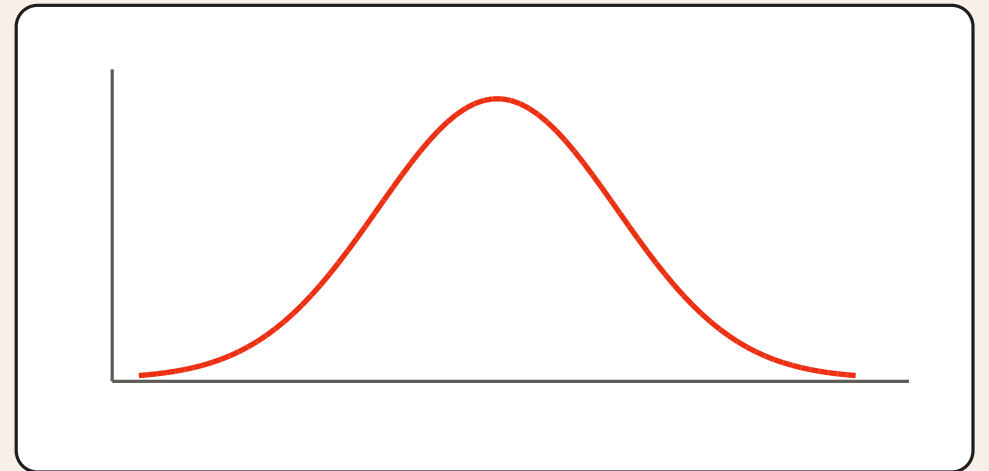


CLASS TOPIC

Sampling distributions and the central limit theorem

How to report

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CLASS TOPIC

Sampling distributions and the central limit theorem

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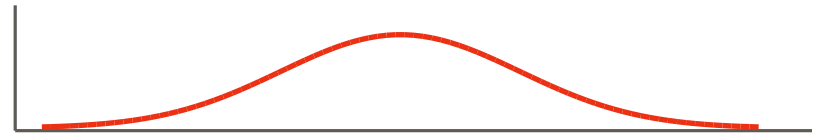
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CLASS TOPIC

Point and interval estimation

What this class fixes

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W09**Statistics**

Clean formulas · aligned layout ·
corrected terminology



CLASS TOPIC

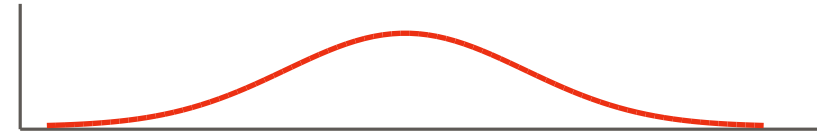
Point and interval estimation

Economic question first

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FORMULA

estimator $\pm$ margin of error
 $CI = \text{estimate} \pm \text{critical value} \times SE$



CLASS TOPIC

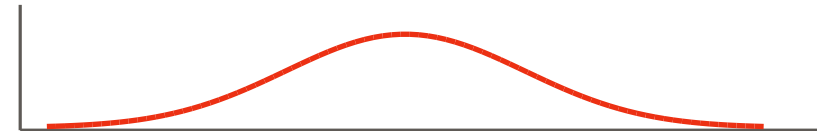
Point and interval estimation

Key definitions

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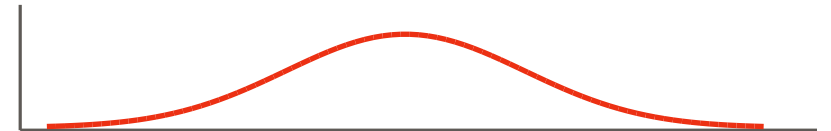
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CLASS TOPIC

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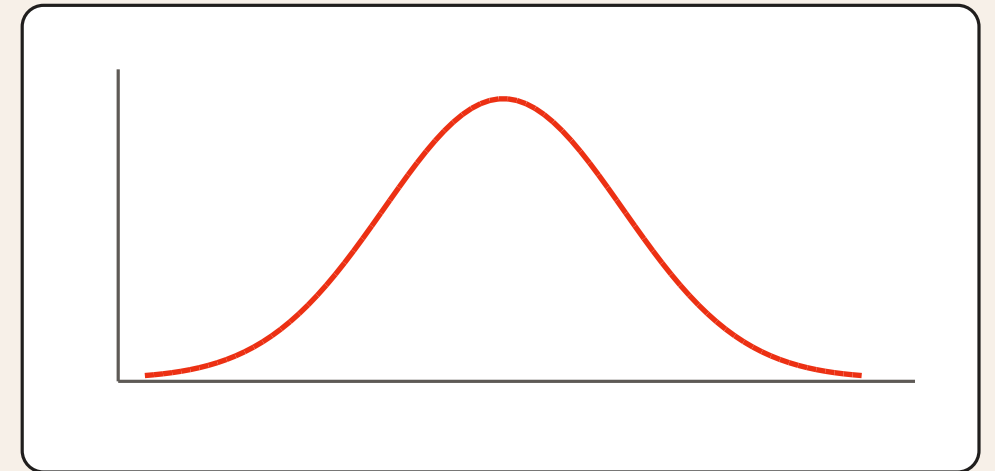
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CLASS TOPIC

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CLASS TOPIC

Point and interval estimation

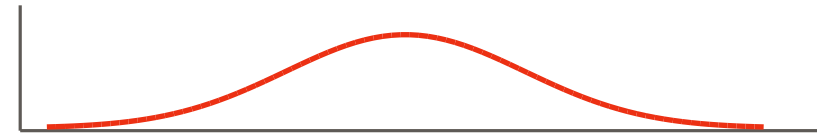
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FORMULA 2

mean: $\bar{x} \pm t^* \cdot s/\sqrt{n}$

proportion: $\hat{p} \pm z^* \sqrt{[\hat{p}(1-\hat{p})/n]}$



CLASS TOPIC

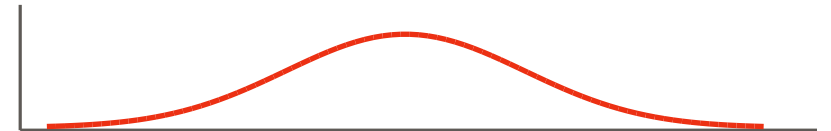
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CLASS TOPIC

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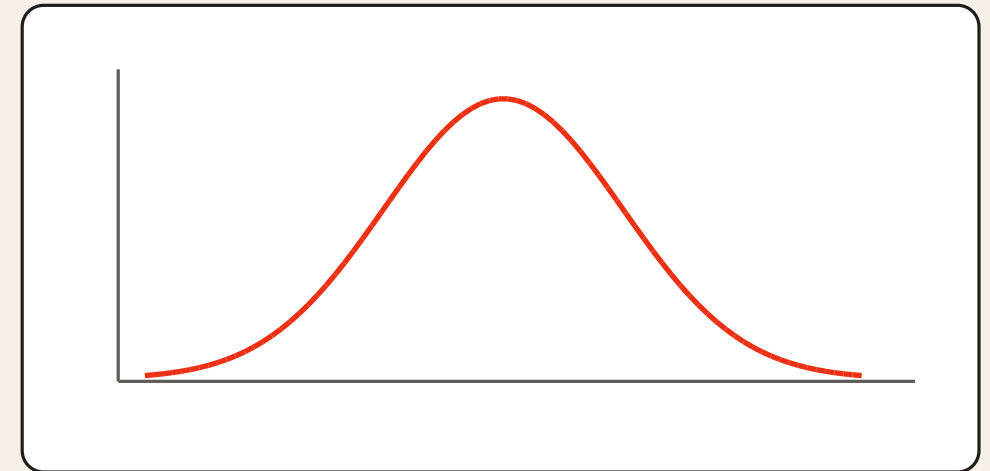
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CLASS TOPIC

Point and interval estimation

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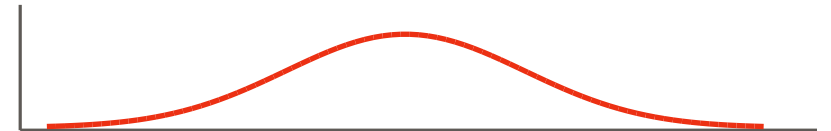
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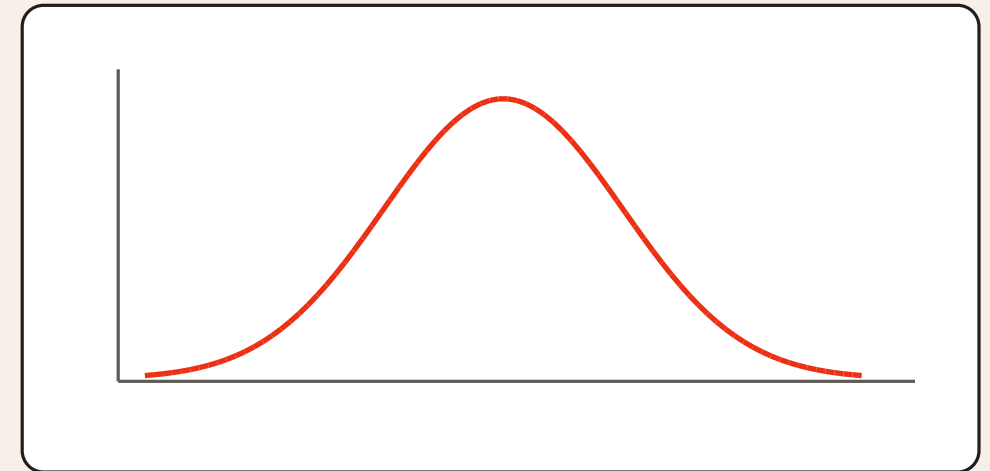


CLASS TOPIC

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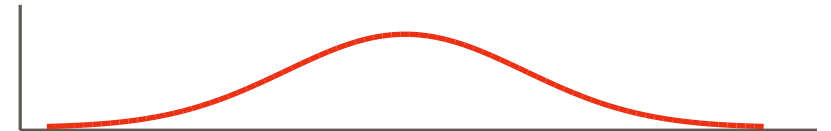
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CLASS TOPIC

Hypothesis testing I - means and proportions

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W10**Statistics**

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CLASS TOPIC

Hypothesis testing I - means and proportions

Economic question first

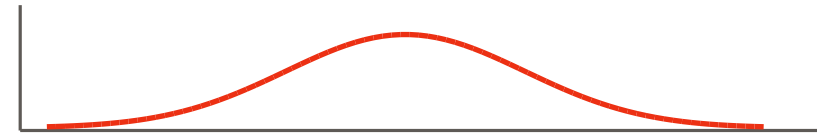
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FORMULA

$$H_0: \mu = \mu_0$$

$$H_1: \mu \neq \mu_0 \text{ (or } >, < \text{)}$$

$$\text{test statistic} = (\text{estimate} - \text{null value}) / \text{SE}$$



CLASS TOPIC

Hypothesis testing I - means and proportions

Key definitions

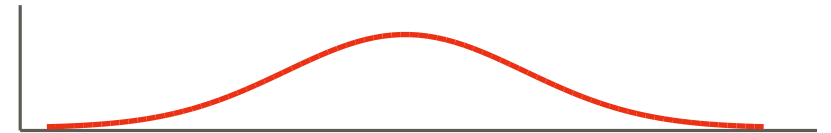
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CLASS TOPIC

Hypothesis testing I - means and proportions

Main formula box

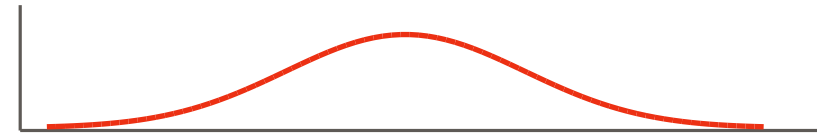
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CLASS TOPIC

Hypothesis testing I - means and proportions

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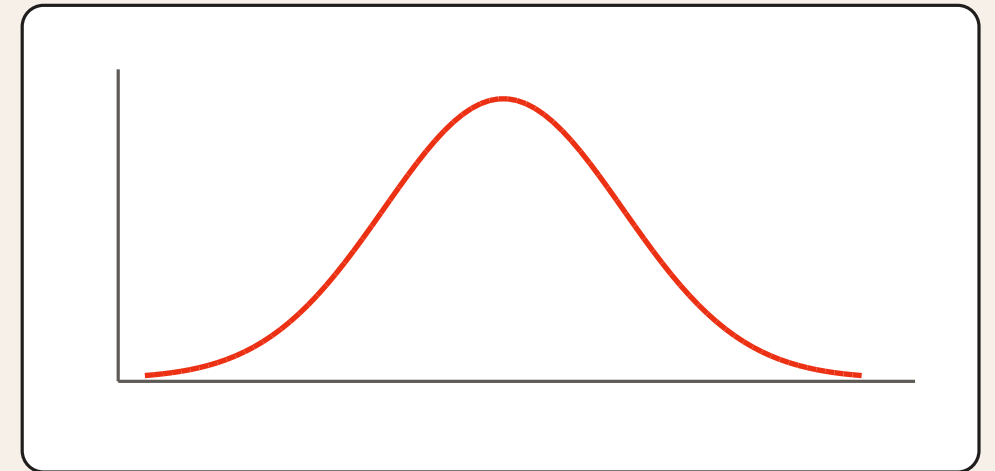
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CLASS TOPIC

Hypothesis testing I - means and proportions

Visual intuition

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CLASS TOPIC

Hypothesis testing I - means and proportions

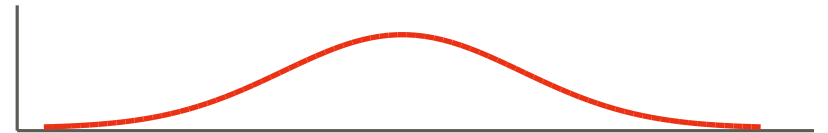
Second formula box

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FORMULA 2

$$t = (\bar{x} - \mu_0)/(s/\sqrt{n})$$

$$z = (\hat{p} - p_0)/\sqrt{[p_0(1-p_0)/n]}$$



CLASS TOPIC

Hypothesis testing I - means and proportions

Common mistakes

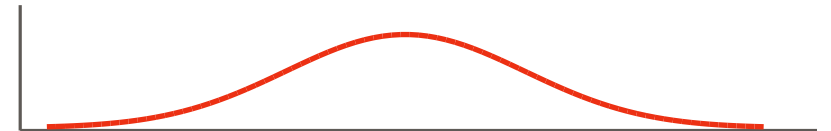
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CLASS TOPIC

Hypothesis testing I - means and proportions

R / spreadsheet check

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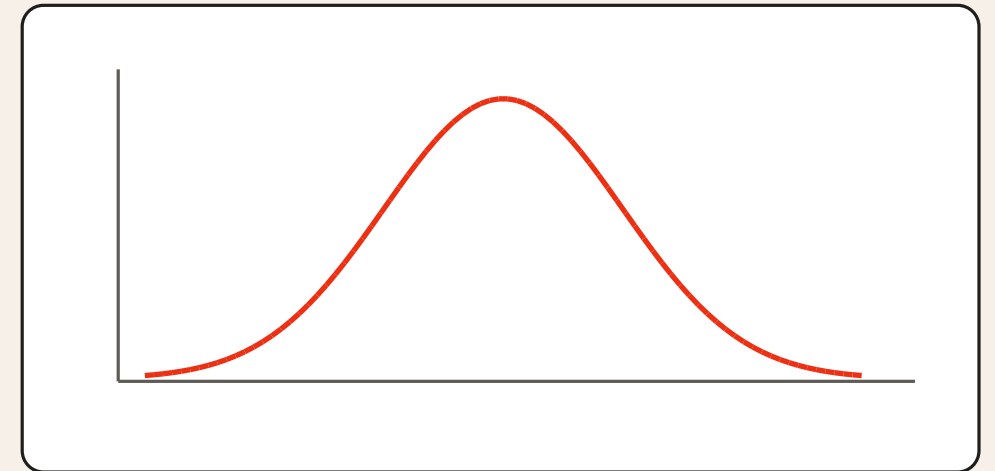
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CLASS TOPIC

Hypothesis testing I - means and proportions

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CLASS TOPIC

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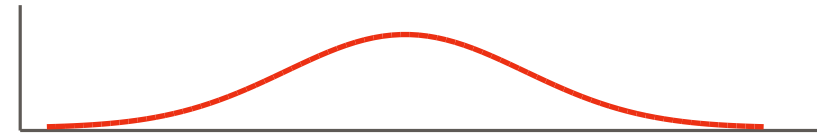
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CLASS TOPIC

Hypothesis testing I - means and proportions

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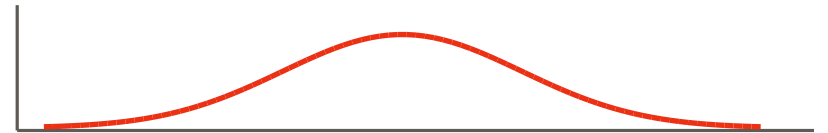
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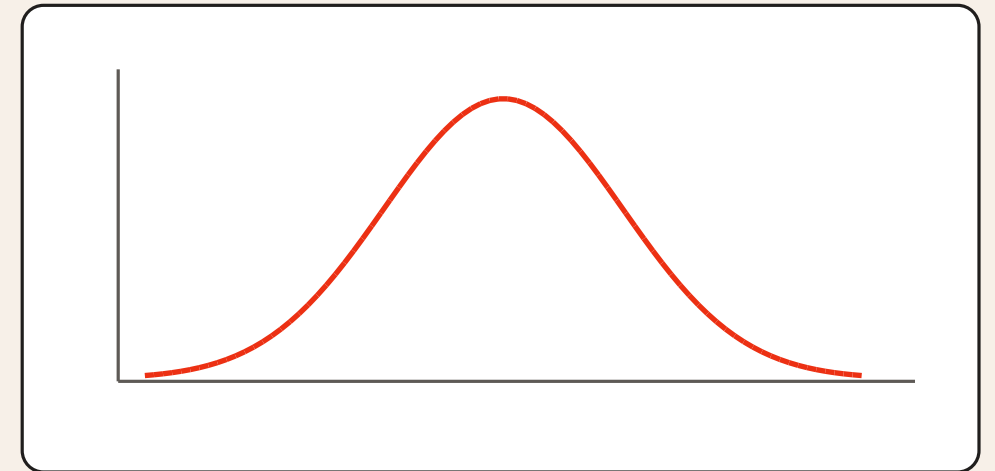


CLASS TOPIC

Hypothesis testing I - means and proportions

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CLASS TOPIC

Hypothesis testing I - means and proportions

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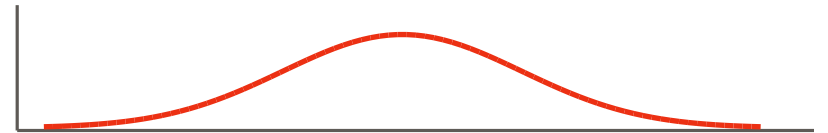
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CLASS TOPIC

Hypothesis testing II - two samples and chi-square

What this class fixes

- Know the central question of the topic.
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W11**Statistics**

Clean formulas · aligned layout ·
corrected terminology



CLASS TOPIC

Hypothesis testing II - two samples and chi-square

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two means: $(\bar{x}_1 - \bar{x}_2) \pm t^* \cdot SE$
 $SE = \sqrt{s_1^2/n_1 + s_2^2/n_2}$



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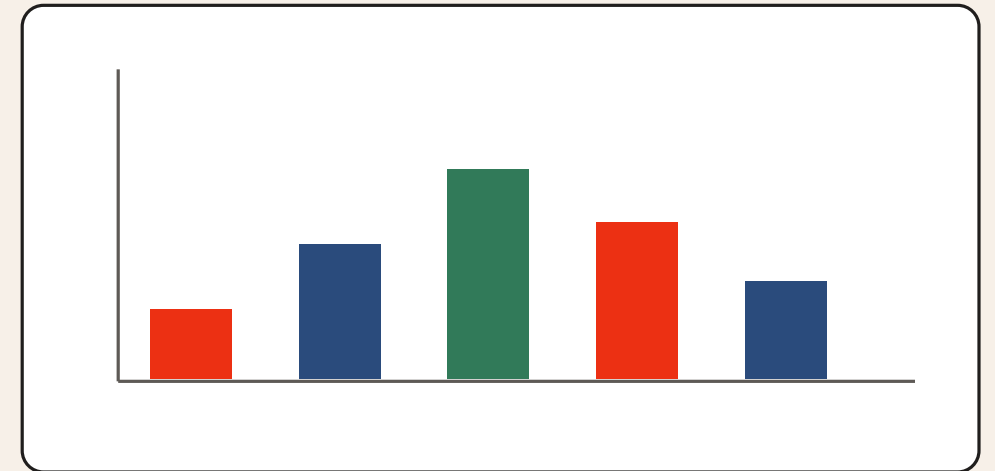
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CLASS TOPIC

Hypothesis testing II - two samples and chi-square

Second formula box

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FORMULA 2

$$\chi^2 = \sum (\text{Observed} - \text{Expected})^2 / \text{Expected}$$
$$\text{df} = (\text{rows} - 1)(\text{columns} - 1)$$



CLASS TOPIC

Hypothesis testing II - two samples and chi-square

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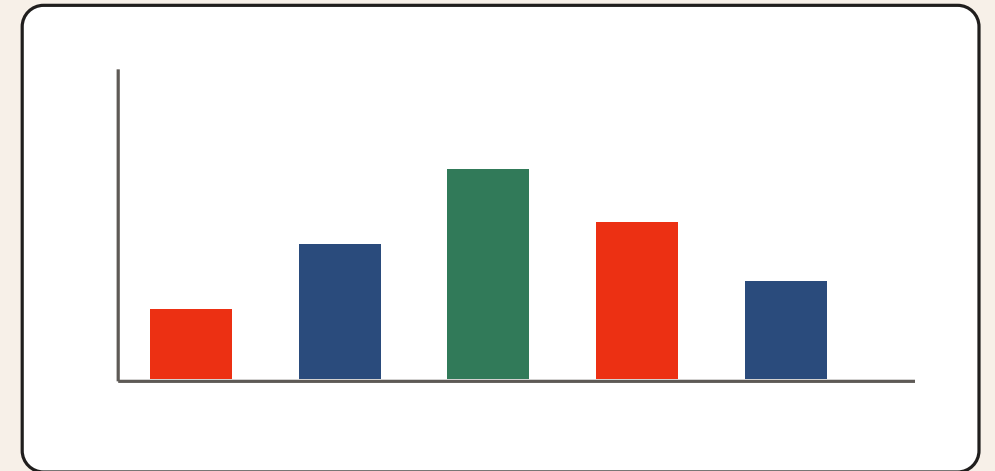
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CLASS TOPIC

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CLASS TOPIC

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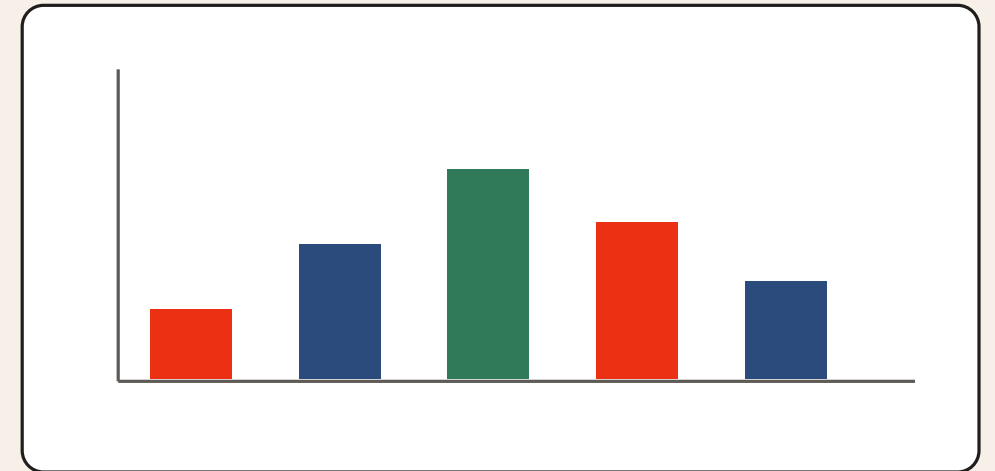


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CLASS TOPIC

Simple linear regression and residual diagnostics

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W12

Statistics

Clean formulas · aligned layout ·
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CLASS TOPIC

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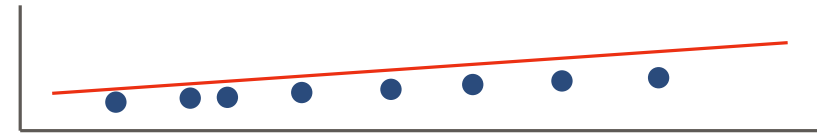
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FORMULA

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$$\text{Estimated line: } \hat{y}_i = b_0 + b_1 x_i$$



CLASS TOPIC

Simple linear regression and residual diagnostics

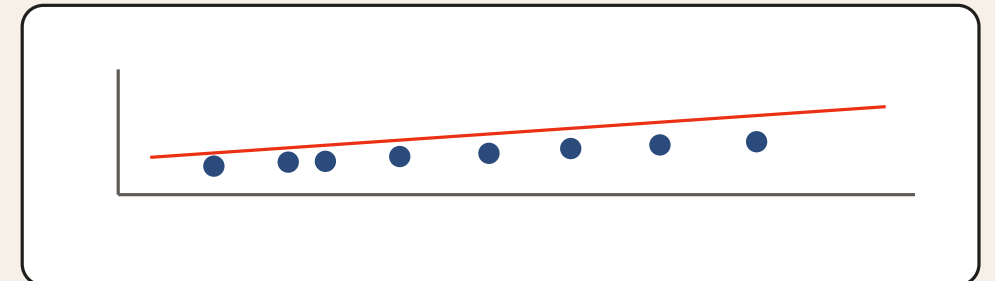
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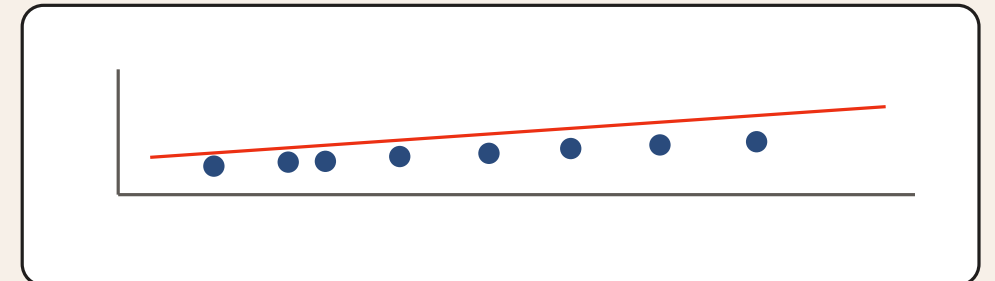
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CLASS TOPIC

Simple linear regression and residual diagnostics

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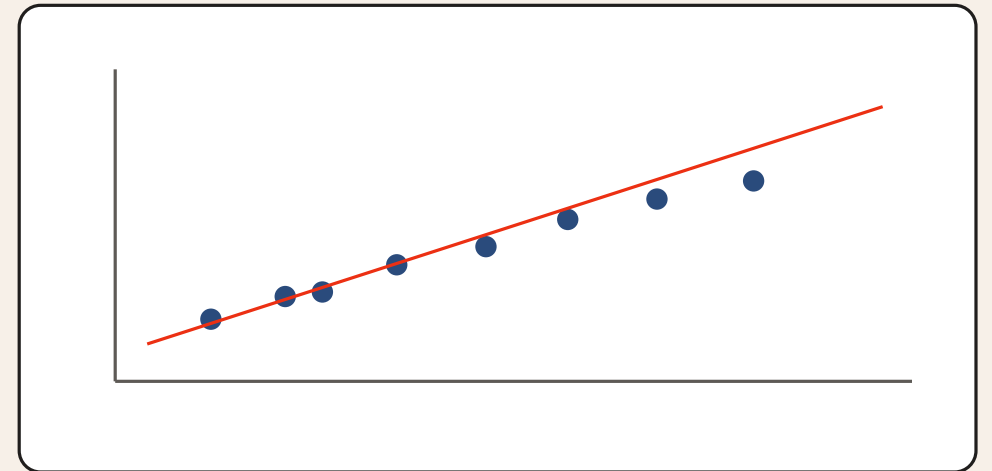
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CLASS TOPIC

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CLASS TOPIC

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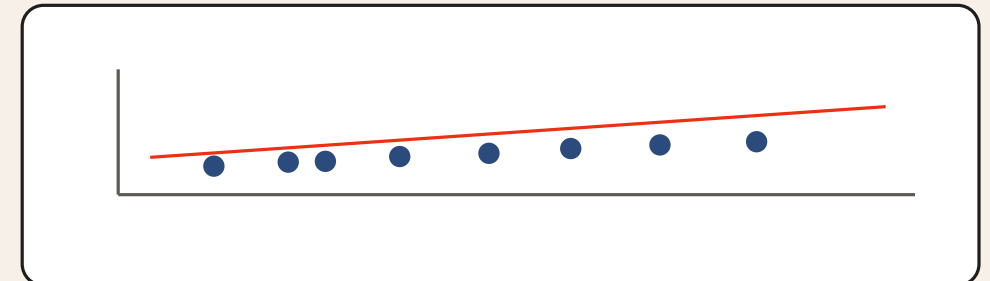
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FORMULA 2

$$e_i = y_i - \hat{y}_i$$

$$SSE = \sum e_i^2$$

OLS chooses b_0 , b_1 to minimize SSE



CLASS TOPIC

Simple linear regression and residual diagnostics

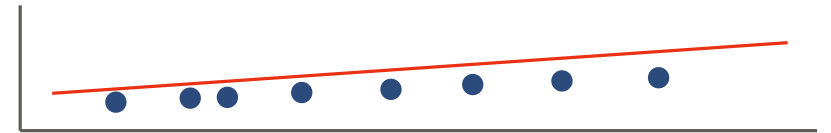
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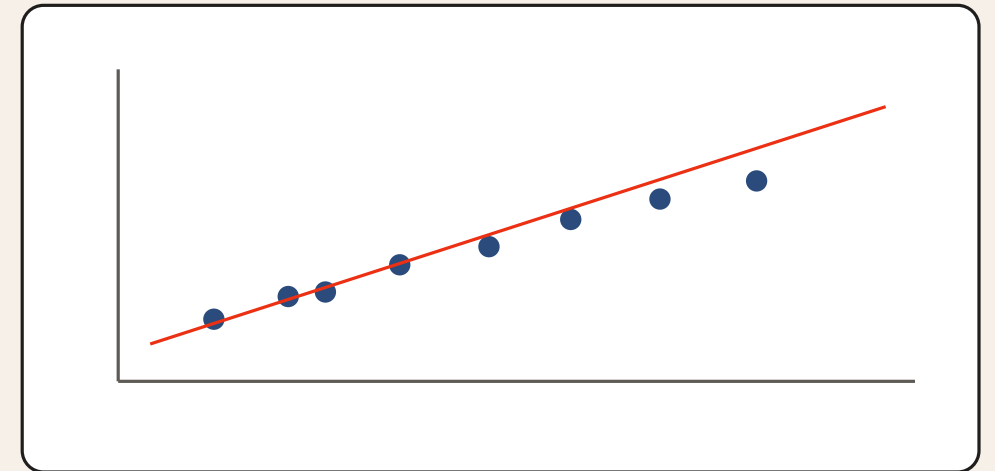
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CLASS TOPIC

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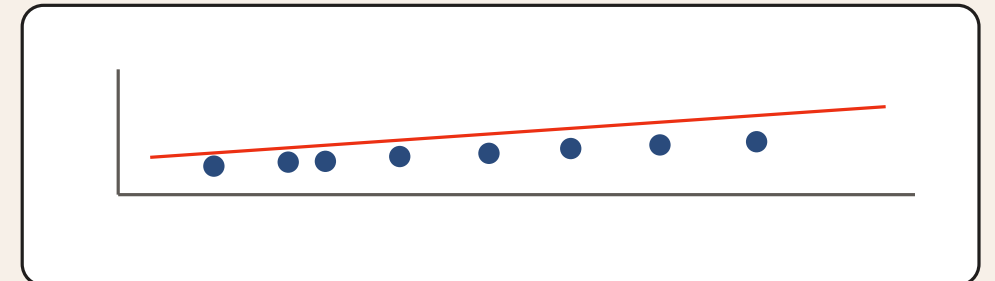
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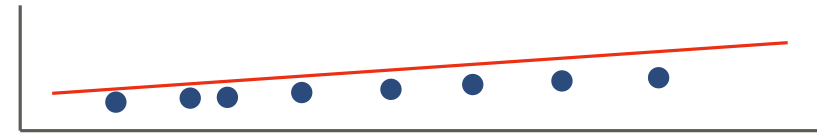
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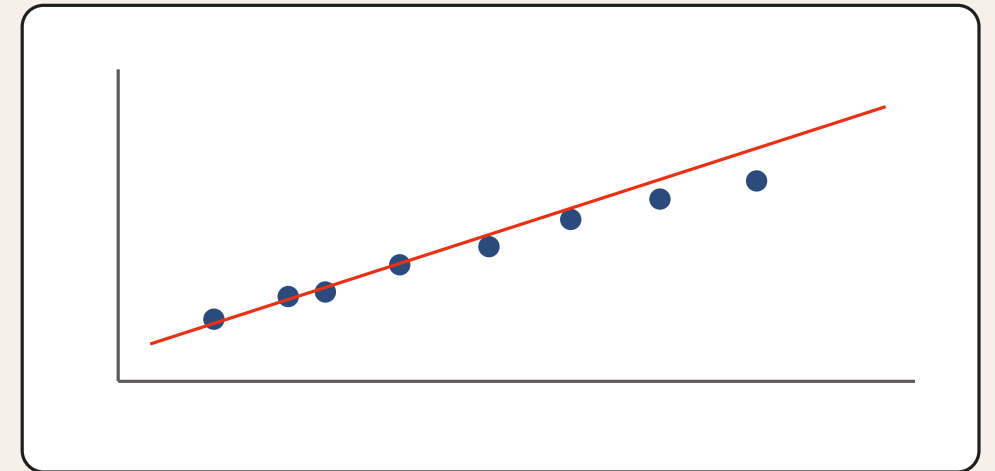


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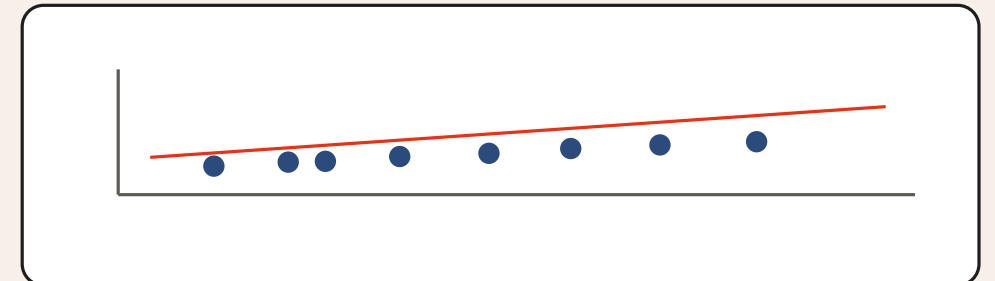
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CLASS TOPIC

Inference in regression; correlation vs causation

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W13**Statistics**

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CLASS TOPIC

Inference in regression; correlation vs causation

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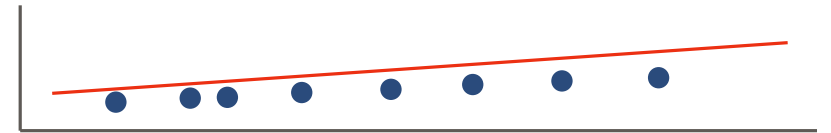
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CLASS TOPIC

Inference in regression; correlation vs causation

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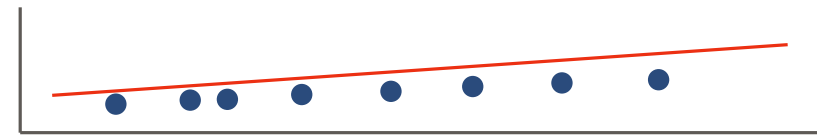
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CLASS TOPIC

Inference in regression; correlation vs causation

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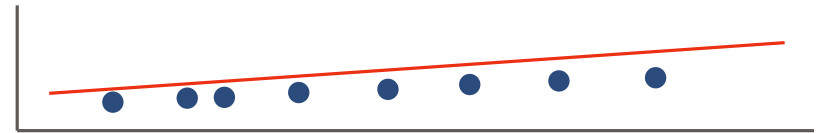
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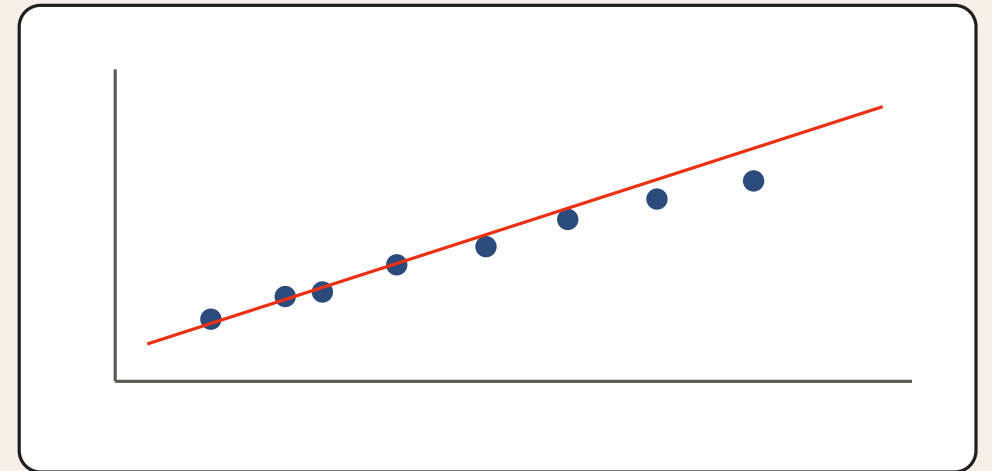
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CLASS TOPIC

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CLASS TOPIC

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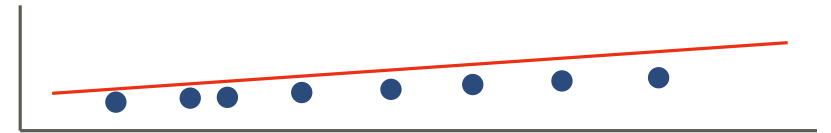
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FORMULA 2

$\text{Corr}(X,Y) \neq \text{causal effect}$

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + u$$

omit $Z \rightarrow$ possible bias in β_1



CLASS TOPIC

Inference in regression; correlation vs causation

Common mistakes

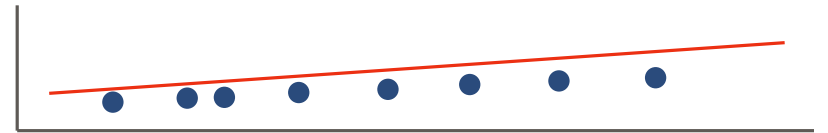
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FORMULA

$$H_0: \beta_1 = 0$$

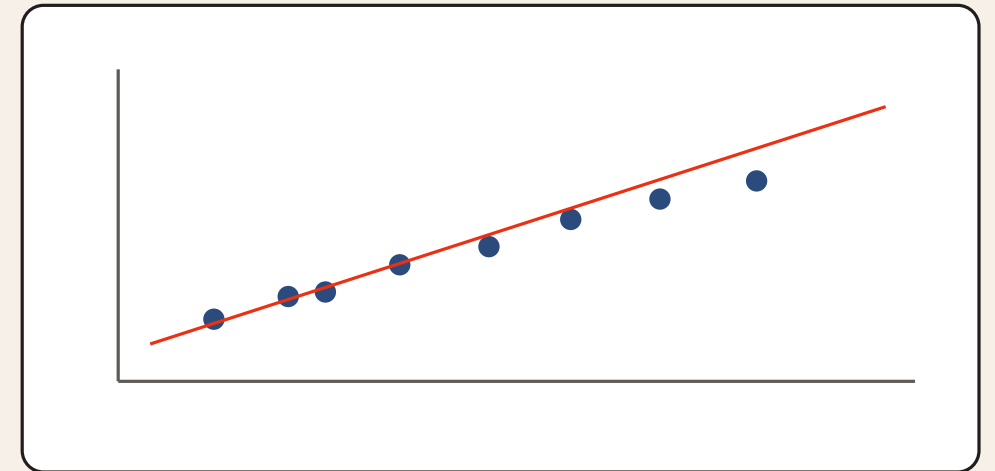
$$t = (b_1 - 0) / SE(b_1)$$

CLASS TOPIC

Inference in regression; correlation vs causation

Interpretation sentence

- Do not stop at the number.
- Explain what the result means in the original context.
- Use cautious language when evidence is limited.

**FORMULA**

$$H_0: \beta_1 = 0$$

CLASS TOPIC

Inference in regression; correlation vs causation

Conditions and assumptions

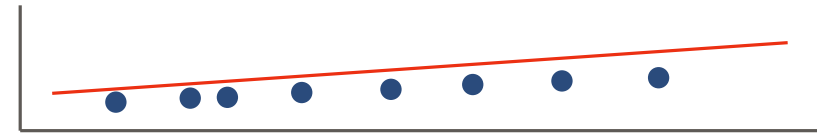
- Check whether the method fits the data type.
- Check independence and sample size when needed.
- If assumptions fail, explain the limitation.

FORMULA

$$H_0: \beta_1 = 0$$

$$t = (b_1 - 0) / SE(b_1)$$

$$CI: b_1 \pm t^*SE(b_1)$$



CLASS TOPIC

Inference in regression; correlation vs causation

Class exercise

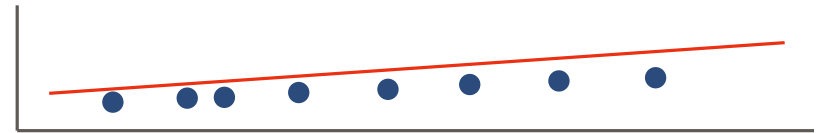
- Solve one calculation by hand.
- Then verify it with software.
- Write a two-sentence conclusion.

FORMULA

$$H_0: \beta_1 = 0$$

$$t = (b_1 - 0) / SE(b_1)$$

$$CI: b_1 \pm t^*SE(b_1)$$

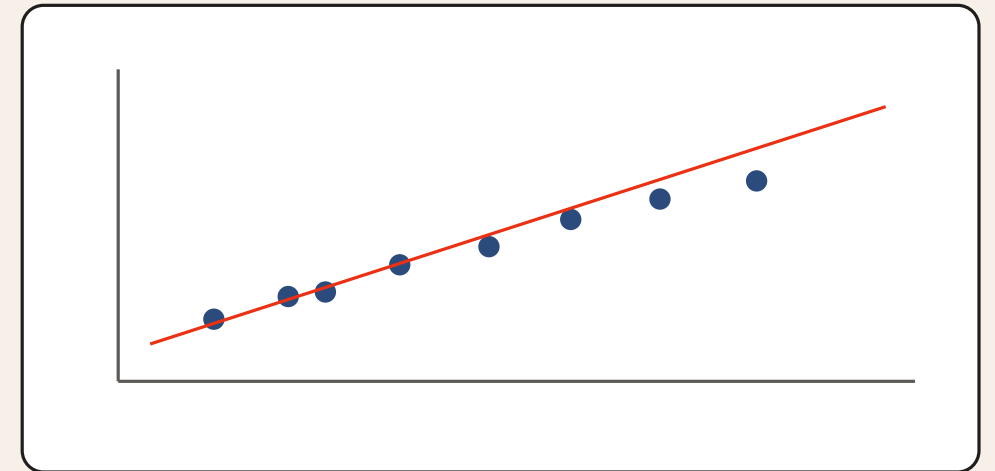


CLASS TOPIC

Inference in regression; correlation vs causation

How to report

- Report method, estimate, uncertainty and interpretation.
- Use tables only when they make comparison easier.
- Separate statistical evidence from causal claims.

**FORMULA**

$$H_0: \beta_1 = 0$$

CLASS TOPIC

Inference in regression; correlation vs causation

Final checklist

- Question is clear.
- Variables and units are defined.
- Formula is correct.
- Result is interpreted.
- Limitations are stated.

FORMULA

$$H_0: \beta_1 = 0$$

$$t = (b_1 - 0) / SE(b_1)$$

$$CI: b_1 \pm t^*SE(b_1)$$

